

IEAP-2026 Series 03 — Data wrangling and ANOVA

RStudio assignment · Team TUBA / DEEMUA / GILLES

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9 October 2026

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About this report

- **GitHub repository (public):** <https://github.com/ChloeGil-student/IEAP-Series03-RStudio>
- **Goal:** reproduce the main result of Mottet et al. (2017) (their Fig. 3: movement time as a function of the index of difficulty, by group) with a full ANOVA pipeline, and discuss why our numbers differ from the published ones.
- **Structure:** this file is the *master document*. Each part is written in its own file in **Sections/** and included here at render time. Section numbers 1.1 to 1.7 are those of the series.

Section	File	Owner
1.1–1.3 Question, data reorganization, ANOVA	Sections/01_Data-reorganisation.qmd	Chloé GILLES
1.4 Linear regression by GROUP	Sections/02_Linear-regression-by-group.qmd	Charles Sam DEEMUA
1.5–1.6 Graph and PDF export	Sections/03_Save-the-graph.qmd	Tuba TUBA
1.7 Discussion of the differences	Sections/04_Discussion-of-the-differences.qmd	Chloé GILLES
2–4 Git workflow, challenges, checklist	Sections/05_GitWorkflow.qmd, Sections/06_Checklist.qmd	All together

Setup

The whole report starts with one command. It runs **R/setup.R**, which loads the libraries, reads the data and builds the clean data set used by every section.

```
source(here::here("R", "setup.R"))

theme_set(theme_classic(base_size = 11)) # same look for all graphs
```

This is the content of **R/setup.R**. The reasons for the cleaning step are given in section 1.3.

```
# R/setup.R
# Shared setup: libraries + data loading + cleaning.
# Sourced once at the top of the master document.
library(ez)
library(tidyverse)
library(here)
```

```
# Load the raw data
Results <- read.table(here("data", "Results.txt"), header = TRUE, sep = " ",
                      stringsAsFactors = TRUE)

# Clean data: remove ID3 and REP3 (incomplete conditions)
clean_results <- Results %>%
  filter(ID != "ID3") %>%
  filter(REP != "R3") %>%
  droplevels() %>%
  mutate(
    GROUP = factor(GROUP, levels = c("pp", "ap", "cp")),
    ID_num = ifelse(ID == "ID1", 3.80, 5.43)
  )
```

1 ANOVA: does the effect of difficulty depend on the group?

1.1 Think before coding

The article (Mottet et al. 2017) compares three groups on a reaching task performed at two levels of difficulty. Its Fig. 3 plots movement time (MT) against the index of difficulty (ID) for each group. Our scientific question is therefore:

Does the effect of task difficulty on movement time depend on the group?

Before writing any code, we fixed what each part of the question means in the data file.

In the question	In Results.txt	Role in the analysis
Movement time	MovementTime (s)	dependent variable
Difficulty	ID (ID1, ID2, ID3)	within -subject factor: a participant does several levels
Group	GROUP (pp, ap, cp)	between -subject factor: a participant belongs to one group
Participant	SUBJ	unit of observation

The article has 19 patients, 13 age-matched controls and 19 young controls. We will check in section 1.3 that the group sizes in the file match. The sizes identify **ap** as the age-matched controls (13 participants), but **pp** and **cp** both have 19. For these two we rely on the participant codes: those of **pp** (ppS01, ppS03, ...) carry the same numbers as the patients P01 to P29 in Table 1 of the article.

The question then splits into the four sub-questions of the series:

1. *Is the effect of difficulty the same in all groups?* This is the **GROUP × ID interaction**, not the two main effects.
2. *How do we visualize it?* With one line per group across the ID levels: parallel lines mean no interaction.

3. *Is there a significant difference?* We test it with a mixed ANOVA (GROUP between, ID within).
4. *What sentence do we write in the report?* See the end of section 1.3.

One point needed attention from the start: the file has one row per reaching movement, but the ANOVA compares participants. Movements of the same person are not independent observations, so they must be averaged per participant before the test.

1.2 Which library?

We use **ez**, as suggested in the series: `ezANOVA()` runs a mixed ANOVA in one call. We looked at what it does with our file: when it receives several rows per participant and condition, it averages them itself and only says so in a warning. We prefer to do this averaging ourselves (section 1.3), so that the step is visible in the report and the ANOVA receives exactly one value per participant and ID level.

1.3 Data reorganization

The data are loaded by `R/setup.R` (see *Setup* at the start of the report). The raw table **Results** has 7560 rows, one per reaching movement.

1.3.1 Looking for missing data

A within-subject factor needs every participant in every level. As in the series, we first count the rows in each GROUP $\times$ ID cell.

```
Results %>%
  count(GROUP, ID) %>%
  pivot_wider(names_from = ID, values_from = n)
```

```
# A tibble: 3 x 4
  GROUP ID1 ID2 ID3
  <fct> <int> <int> <int>
1 ap     740   740   740
2 cp    1140  1140  1140
3 pp     760   760   400
```

The group **pp** has about half the expected rows at ID3. Counting rows shows that something is missing, but not who. Counting participants answers both questions at once, for ID and for the repetition REP:

```
Results %>%
  group_by(GROUP, ID, REP) %>%
  summarise(participants = n_distinct(SUBJ), .groups = "drop") %>%
  pivot_wider(names_from = REP, values_from = participants)
```

```
# A tibble: 9 x 5
  GROUP ID      R1    R2    R3
  <fct> <fct> <int> <int> <int>
1 ap    ID1     13    13    11
2 ap    ID2     13    13    11
3 ap    ID3     13    13    11
4 cp    ID1     19    19    19
5 cp    ID2     19    19    19
6 cp    ID3     19    19    19
7 pp    ID1     19    19    NA
8 pp    ID2     19    19    NA
9 pp    ID3     10    10    NA
```

The group sizes are 13, 19 and 19, as in the article. Two conditions are incomplete:

- **ID3** was done by only 10 of the 19 patients. Keeping it would mean either losing 9 patients or comparing the groups on different conditions.
- **R3** was done by no patient and by 11 of the 13 age-matched controls.

1.3.2 Two decisions

We also remove R3 for everybody. This one is a choice, not an obligation: the ANOVA would run with R3, because the rows are averaged per participant anyway. Our main reason is the design of the article: two levels of difficulty and 80 reaching movements per participant (Mottet et al. 2017, Methods), that is two repetitions in each of the two trials. R3 is an extra repetition that the article does not analyse. Without ID3 and R3, our data have exactly the structure of the article (checked below). We also looked at the repetitions themselves:

```
Results %>%
  filter(ID != "ID3") %>%
  group_by(GROUP, REP) %>%
  summarise(mean_MT = round(mean(MovementTime), 2), .groups = "drop") %>%
  pivot_wider(names_from = REP, values_from = mean_MT)
```

```
# A tibble: 3 x 4
  GROUP    R1    R2    R3
  <fct> <dbl> <dbl> <dbl>
1 ap    1.17  1.1  1.13
2 cp    0.99  0.88  0.89
3 pp    4.22  3.55 NA
```

The first repetition is slower in the three groups, so with R3 the average of a control participant would include one more practised repetition than the average of a patient. The effect on our comparison would be small, however: for the controls R3 is almost equal to R2, and the groups differ by 2 to 3 s. Practice is a secondary reason. Since no patient did R3, removing it cannot change the results of the patients either, which matters for section 1.7.

1.3.3 The cleaning rule

Both decisions are written once, in the `filter()` line of `R/setup.R` (shown in *Setup*), so that every section works on the same `clean_results`.

As in the series, we recount after cleaning. Every cell is now filled, and within a group all cells have the same number of rows:

```
clean_results %>%  
  count(GROUP, ID, REP) %>%  
  pivot_wider(names_from = REP, values_from = n)
```

```
# A tibble: 6 x 4  
  GROUP ID      R1    R2  
  <fct> <fct> <int> <int>  
1 pp    ID1      380   380  
2 pp    ID2      380   380  
3 ap    ID1      260   260  
4 ap    ID2      260   260  
5 cp    ID1      380   380  
6 cp    ID2      380   380
```

The cleaning keeps 4080 of the 7560 rows (54 %). The price is that we set aside valid data, mostly from the two control groups.

1.3.4 One value per participant and ID level

```
subject_means <- clean_results %>%  
  group_by(GROUP, SUBJ, ID, ID_num) %>%  
  summarise(n_movements = n(),  
            MovementTime = mean(MovementTime),  
            .groups = "drop")  
  
# Safety check: every participant must have both ID levels  
stopifnot(all(count(subject_means, SUBJ)$n == 2))
```

We check the result of the cleaning on this table: number of participants per group, and number of movements behind each participant mean.

```
subject_means %>%  
  group_by(GROUP) %>%  
  summarise(participants = n_distinct(SUBJ),  
            min_movements = min(n_movements),  
            max_movements = max(n_movements))
```

```
# A tibble: 3 x 4
  GROUP participants min_movements max_movements
  <fct>         <int>         <int>         <int>
1 pp             19             40             40
2 ap             13             40             40
3 cp             19             40             40
```

Every participant mean is based on 40 movements, so the design is balanced within participants, and 2×40 gives the 80 movements of the article. The groups, however, do not have the same number of participants: the design is not balanced between groups. Cleaning cannot fix that, and it is the reason for `type = 3` in the ANOVA below.

1.3.5 Visualize the interaction

We draw the interaction plot of the series: the mean movement time in each `GROUP × ID` cell, with error bars of half a standard deviation on each side, computed over all the movements of the cell.

```
ggplot(clean_results) +
  aes(x = ID, y = MovementTime, group = GROUP,
      color = fct_relevel(GROUP, "ap", "cp")) + # group order of the series
  stat_summary(fun = mean, geom = "line") +
  stat_summary(fun = mean, geom = "errorbar", width = 0.01,
              fun.min = \(x) mean(x) - sd(x) / 2,
              fun.max = \(x) mean(x) + sd(x) / 2) +
  stat_summary(fun = mean, geom = "point", shape = 21, fill = "white", size = 3) +
  labs(title = "Interaction between GROUP and ID on MovementTime",
       x = "ID", y = "Mean MovementTime (s)", color = "GROUP") +
  theme_minimal()
```

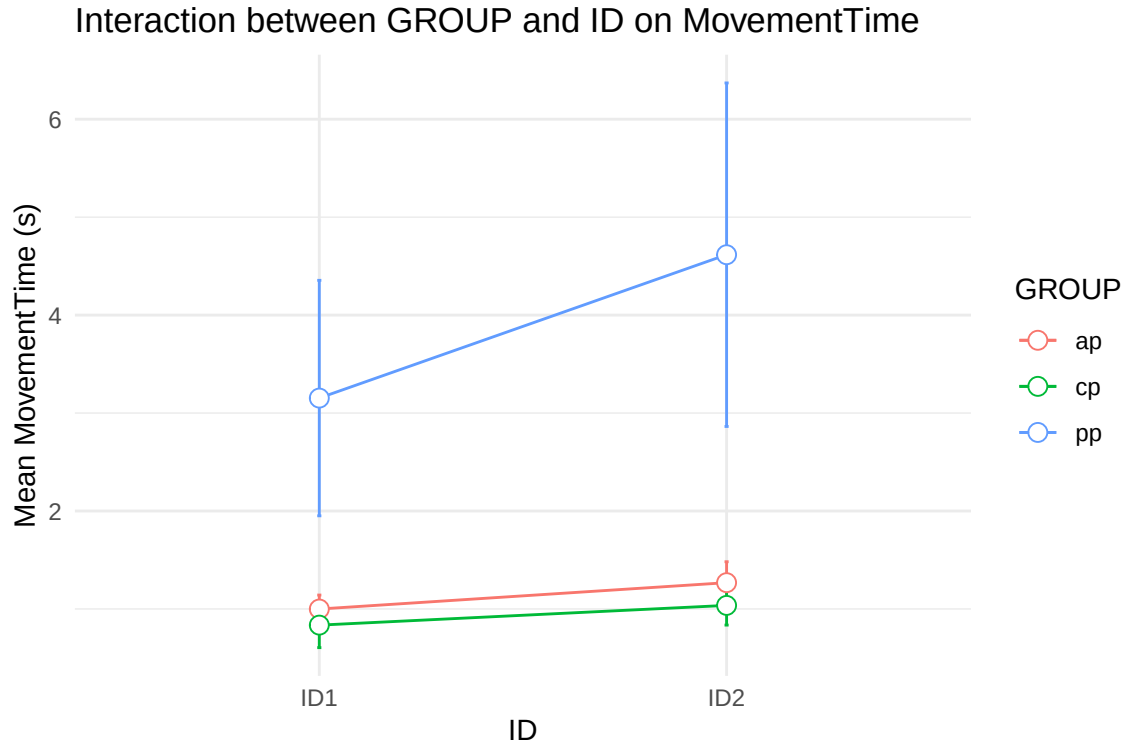


Figure 1: Interaction between GROUP and ID. Points: mean of all movements of the cell; bars: plus or minus half a standard deviation of these movements.

The lines are not parallel in Figure 1: movement time rises much more between ID1 and ID2 for the patients than for the two control groups.

The error bars of this plot describe the spread of individual movements. The ANOVA, however, works on one value per participant, so what matters for the test is the spread between participants. The table below gives the mean and the standard deviation of the participant means in each cell. The means are the same as in the plot, because every participant has the same number of movements; the standard deviations are not.

```
cell_means <- subject_means %>%
  group_by(GROUP, ID, ID_num) %>%
  summarise(n = n(),
            mean = mean(MovementTime),
            sd = sd(MovementTime),
            .groups = "drop")

knitr::kable(cell_means, digits = 2)
```

GROUP	ID	ID_num	n	mean	sd
pp	ID1	3.80	19	3.15	1.47
pp	ID2	5.43	19	4.62	2.41

GROUP	ID	ID_num	n	mean	sd
ap	ID1	3.80	13	1.00	0.20
ap	ID2	5.43	13	1.27	0.29
cp	ID1	3.80	19	0.83	0.12
cp	ID2	5.43	19	1.04	0.11

This table shows a problem for the ANOVA, which assumes similar variances in all groups: the largest standard deviation is 22 times the smallest. We come back to this in section 1.7.

1.3.6 Run the ANOVA

```
res_anova <- ezANOVA(
  data      = as.data.frame(subject_means), # ez expects a plain data frame
  dv        = MovementTime,
  wid        = SUBJ,
  between    = GROUP,
  within     = ID,
  type      = 3 # groups of unequal size
)
```

Warning: Data is unbalanced (unequal N per group). Make sure you specified a well-considered value for the type argument to ezANOVA().

```
res_anova$ANOVA
```

	Effect	DFn	DFd	F	p	p<.05	ges
2	GROUP	2	48	37.24408	1.720007e-10	*	0.57462110
3	ID	1	48	26.10886	5.543011e-06	*	0.06581307
4	GROUP:ID	2	48	11.58124	7.866206e-05	*	0.05882293

Warnings are hidden everywhere else in this report, but not for this chunk: `ezANOVA()` reminds us that the groups are of unequal size, which is what we saw above. There is no sphericity test in the output because ID has only two levels. The table is identical to the one in the series, which confirms that averaging per participant ourselves is what `ezANOVA()` was doing silently.

1.3.7 The sentence for the report

Movement time differed between the groups (main effect of GROUP, $F(2, 48) = 37.24$, $p < .001$) and was longer for the more difficult task (main effect of ID, $F(1, 48) = 26.11$, $p < .001$). **The effect of difficulty was not the same in all groups** (GROUP $\times$ ID interaction, $F(2, 48) = 11.58$, $p < .001$): the increase of movement time with difficulty was steeper for the patients (Figure 1).

The effect sizes put this result in perspective. The generalized eta squared (column `ges`) is 0.57 for GROUP but only 0.06 for the interaction. The main difference between the groups is that patients are slower at *both* levels of difficulty; the fact that difficulty costs them more is real, but it comes second.

The ANOVA says that the groups differ, not which ones. We did not run post-hoc comparisons here; they are in section 1.7, together with a check of the ANOVA that does not assume equal variances.

1.4 Linear regression by GROUP

The ANOVA treats ID1 and ID2 as two labels. The article goes one step further and describes each group by a line, $MT = a + b \times ID$ (Fitts's law), where ID is a number of bits, the slope b is the time cost of one more bit of difficulty and the intercept a is the predicted time for a movement without accuracy constraint (Mottet et al. 2017). We compute this line for each group.

1.4.1 ID as a number

A regression needs a numerical predictor. In the article, the two levels of difficulty are 3.80 and 5.43 bits (Mottet et al. 2017, Methods). The conversion is done once, in `R/setup.R` (column `ID_num`, shown in *Setup*). We check that each label received the right value:

```
clean_results %>%  
  count(ID, ID_num)
```

	ID	ID_num	n
1	ID1	3.80	2040
2	ID2	5.43	2040

1.4.2 One regression per group

The function below fits the line for each group of a data frame and returns the intercept, the slope and the equation as text. We wrote a function because we fit the same model on other versions of the data in section 1.7.

```
fit_by_group <- function(data) {  
  data %>%  
    group_by(GROUP) %>%  
    group_modify(~ {  
      model <- lm(MovementTime ~ ID_num, data = .x)  
      tibble(intercept = coef(model)[[1]], slope = coef(model)[[2]])  
    }) %>%  
    ungroup() %>%  
    mutate(equation = sprintf("MT = %.3f + %.3f * ID", intercept, slope))  
}
```

```
regression_by_group <- fit_by_group(clean_results)

knitr::kable(regression_by_group, digits = 3)
```

GROUP	intercept	slope	equation
pp	-0.259	0.898	MT = -0.259 + 0.898 * ID
ap	0.372	0.165	MT = 0.372 + 0.165 * ID
cp	0.366	0.123	MT = 0.366 + 0.123 * ID

These are the three equations given in the series, so our cleaning and our conversion of ID reproduce the expected result.

1.4.3 What these equations do and do not tell us

What they tell us. The slope is positive in the three groups: movement time increases with difficulty. One more bit costs 0.9 s to the patients, against 0.17 s and 0.12 s to the age-matched and young controls, that is 5.4 and 7.3 times more. The difference between the slopes is the $\text{GROUP} \times \text{ID}$ interaction of section 1.3, expressed in seconds per bit.

What they do not tell us. We see three limits.

- *The regression adds no information to the two means.* With only two values of ID, the least-squares line of a group passes exactly through its two mean movement times: the slope is $(\text{mean at ID2} - \text{mean at ID1}) / (5.43 - 3.80)$. We cannot test whether the relation is really linear, a limit that the article also states. The file does contain a third level, but not all patients did it and we do not know its value in bits.
- *The intercept is an extrapolation.* It is the predicted movement time at $\text{ID} = 0$, which is 3.8 bits away from the nearest measurement, so a small change in the two means is amplified. The negative intercept of the patients cannot be a movement time; it only says where the extended line crosses the vertical axis.
- *We report coefficients, not their uncertainty.* The model is fitted on all movements as if they were independent, although they come from 51 participants. This does not change the intercepts and slopes, but the standard errors and p-values that `lm()` would print are too optimistic, so we do not report them. Our test of the difference between groups is the ANOVA, which is based on participants.

1.5 Graph of movement time as a function of ID

The goal is a graph that resembles Fig. 3 (left panel) of the article: for each group, the mean movement time at the two levels of difficulty, error bars, the regression line and its equation.

Everything we need already exists: the means and standard deviations are in `cell_means` (section 1.3) and the lines and equations in `regression_by_group` (section 1.4). This section only draws.

1.5.1 Choices we made

- **Error bars.** The caption of Fig. 3 says that the bars are standard deviations between participants. We use the `sd` column of `cell_means`.
- **Lines.** The regression lines are drawn over the whole x axis, as in the article.
- **Axes.** The x axis starts at 0 so that the intercepts are visible, and the y axis goes below 0 because the line of the patients does.
- **Equations.** Their text comes from `regression_by_group`, so the graph cannot disagree with the table of section 1.4.

```
# Position of the three equations, one under the other
equation_labels <- regression_by_group %>%
  mutate(x = 0.3, y = 6.9 - 0.4 * row_number())

graph_MT_ID <- ggplot(cell_means) +
  aes(x = ID_num, y = mean, color = GROUP) +
  geom_abline(
    data = regression_by_group,
    aes(intercept = intercept, slope = slope, color = GROUP),
    linetype = "dashed",
    show.legend = FALSE
  ) +
  geom_errorbar(
    aes(ymin = mean - sd, ymax = mean + sd),
    width = 0
  ) +
  geom_point(
    shape = 21,
    fill = "white",
    size = 3
  ) +
  geom_text(
    data = equation_labels,
    aes(x = x, y = y, label = equation),
    hjust = 0,
    show.legend = FALSE
  ) +
  scale_x_continuous(breaks = 0:6) +
  coord_cartesian(
    xlim = c(0, 6.2),
    ylim = c(-1, 7.5),
    expand = FALSE
  ) +
  labs(
    x = "ID (bit)",
    y = "Movement time (s)"
  )
```

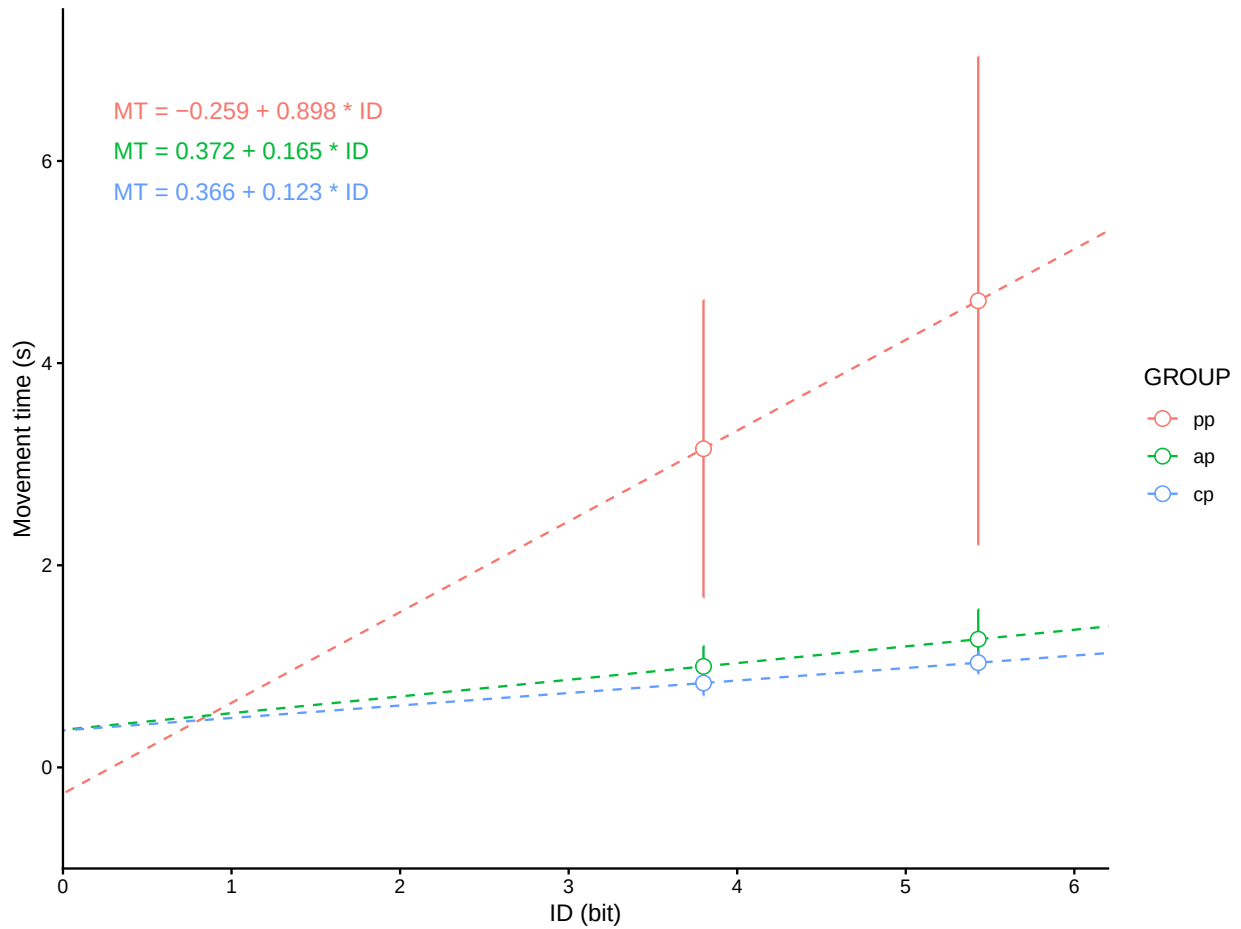


Figure 2: Movement time as a function of the index of difficulty. Points: group means; bars: one standard deviation between participants; dashed lines: linear regression of each group.

1.5.2 Reading the graph

Figure 2 has the same shape as Fig. 3 of the article: two almost flat and almost identical lines for the control groups, and a much higher and steeper line for the patients, with error bars that are several times longer. Two differences are visible without any calculation. Our points for the patients are higher, especially at the more difficult level, and our line for the patients crosses the vertical axis below zero, whereas the three lines of the article start close together around half a second. Section 1.7 looks for the reason.

1.6 Export the graph as a PDF

The journal asks for a figure of 8×6 inches. The file is written in the folder **figures/** of the repository.

```
dir.create(here("figures"), showWarnings = FALSE)

ggsave(here("figures", "MT_as_function_of_ID.pdf"),
       plot = graph_MT_ID, width = 8, height = 6, units = "in")
```

1.7 Discussion: differences with the article

The series tells us that the method of the article was more sophisticated than ours and not fully disclosed. We cannot redo that method, so we ask three questions that our data can answer: how large the differences are, how well our own equations are determined, and where in the data the differences come from. We then check the ANOVA of section 1.3 with tests that need fewer assumptions, and conclude.

1.7.1 How large are the differences?

Comparing intercepts is misleading, because they are extrapolations (section 1.4). We compare what the two sets of equations predict where the data are, at 3.80 and 5.43 bits. The equations of the article are typed from its Fig. 3.

```
article <- tibble(
  GROUP      = factor(c("pp", "ap", "cp"), levels = levels(clean_results$GROUP)),
  intercept  = c(0.422, 0.622, 0.475),
  slope      = c(0.672, 0.118, 0.097)
)

bind_rows("Article (Fig. 3)" = article,
         "Ours"              = select(regression_by_group, GROUP, intercept, slope),
         .id = "source") %>%
  mutate(MT_easy = intercept + slope * 3.80,
         MT_hard  = intercept + slope * 5.43) %>%
  arrange(GROUP) %>%
  select(GROUP, source, intercept, slope, MT_easy, MT_hard) %>%
  knitr::kable(digits = c(0, 0, 3, 3, 2, 2),
               col.names = c("Group", "Source", "Intercept (s)", "Slope (s/bit)",
                             "MT at 3.80 bits (s)", "MT at 5.43 bits (s)"))
```

Group	Source	Intercept (s)	Slope (s/bit)	MT at 3.80 bits (s)	MT at 5.43 bits (s)
pp	Article (Fig. 3)	0.422	0.672	2.98	4.07
pp	Ours	-0.259	0.898	3.15	4.62
ap	Article (Fig. 3)	0.622	0.118	1.07	1.26
ap	Ours	0.372	0.165	1.00	1.27
cp	Article (Fig. 3)	0.475	0.097	0.84	1.00
cp	Ours	0.366	0.123	0.83	1.04

For the two control groups, the article and our analysis predict movement times that differ by less than 0.1 s, even though the intercepts look different (0.62 against 0.37 s for the age-matched group). For the patients, our movement times are longer at both levels, and the gap is about three times larger at the difficult level. This is why our slope is steeper and, by extrapolation, our intercept negative. The disagreement is about the patients.

The ANOVA gives a similar picture:

```
tibble(
  Effect      = c("GROUP", "ID", "GROUP:ID"),
  F_article   = c(35.32, 5.42, 10.02),
  F_ours      = a[Effect, "F"],
  ges_article = c(0.67, 0.03, 0.03),
  ges_ours    = a[Effect, "ges"]
) %>%
  knitr::kable(digits = 2)
```

Effect	F_article	F_ours	ges_article	ges_ours
GROUP	35.32	37.24	0.67	0.57
ID	5.42	26.11	0.03	0.07
GROUP:ID	10.02	11.58	0.03	0.06

The F values of GROUP and of the interaction are close. For ID we cannot compare: the article reports $F(1, 48) = 5.42$ with $p = 0.0000$, but this F value corresponds to $p = 0.024$, so one of the two numbers is probably a typing error.

The 48 degrees of freedom are useful for another reason. With three groups they correspond to 51 participants, so the article kept all 19 patients. Whatever explains the different equations, it is not that some participants were left out.

1.7.2 How well do we know our own equations?

In section 1.4 we did not report the uncertainty of the coefficients, because `lm()` on all movements treats them as independent. We can get it another way. With two levels of difficulty, each participant has their own line through their two mean movement times:

```
participant_lines <- subject_means %>%
  select(GROUP, SUBJ, ID, MovementTime) %>%
  pivot_wider(names_from = ID, values_from = MovementTime) %>%
  mutate(mean_MT = (ID1 + ID2) / 2,
         difference = ID2 - ID1,
         ratio = ID2 / ID1,
         slope = difference / (5.43 - 3.80),
         intercept = ID1 - slope * 3.80)
```

Because every participant has the same number of movements, the mean of the individual slopes is the slope of the group in section 1.4. This also answers a question we had at the start, whether averaging per participant before the regression could explain the difference with the article. It cannot:

```
mean_slopes <- participant_lines %>%
  group_by(GROUP) %>%
  summarise(slope = mean(slope))

all.equal(mean_slopes$slope, regression_by_group$slope)
```

```
[1] TRUE
```

The individual lines give us what was missing, the spread between participants, and therefore a 95 % confidence interval for each coefficient (Student's t on the participants of a group).

```
# mean and 95 % confidence interval as text, e.g. "0.90 [0.47, 1.32]"
mean_ci <- function(x) {
  half <- qt(0.975, length(x) - 1) * sd(x) / sqrt(length(x))
  sprintf("%.2f [%.2f, %.2f]", mean(x), mean(x) - half, mean(x) + half)
}

participant_lines %>%
  group_by(GROUP) %>%
  summarise(n = n(),
            intercept_ours = mean_ci(intercept),
            slope_ours      = mean_ci(slope)) %>%
  left_join(article, by = "GROUP") %>%
  select(GROUP, n, intercept_ours, intercept_article = intercept,
         slope_ours, slope_article = slope) %>%
  knitr::kable(digits = 3,
               col.names = c("Group", "n", "Our intercept [95 % CI]", "Article",
                             "Our slope [95 % CI]", "Article"))
```

Group	n	Our intercept [95 % CI]	Article	Our slope [95 % CI]	Article
pp	19	-0.26 [-1.76, 1.25]	0.422	0.90 [0.47, 1.32]	0.672
ap	13	0.37 [0.21, 0.53]	0.622	0.17 [0.12, 0.21]	0.118
cp	19	0.37 [0.24, 0.50]	0.475	0.12 [0.10, 0.15]	0.097

For the patients the intervals are wide: the intercept could be anywhere within about 1.5 s on either side of zero, and the slope is known to within a factor of almost three. The values of the article are inside both intervals. Our negative intercept is therefore not a result. With 19 patients, the data do not say whether the intercept is negative or positive.

The article does the same thing as we do here: it treats the slope and the intercept as values of each participant and compares the intercepts between groups, without finding a difference ($F(2, 48) = 0.17$, $p = 0.84$). The same test on our individual intercepts:

```
summary(aov(intercept ~ GROUP, data = participant_lines))
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
GROUP	2	4.69	2.346	0.634	0.535
Residuals	48	177.72	3.702		

Our intercepts do not differ between groups either. On this point the two analyses agree, even though the patient intercepts have opposite signs.

For the controls it is the opposite. The intervals are narrow, and the intercept of the article for the age-matched group is outside ours, although the two equations predict the same movement times within 0.1 s. A small difference is visible there because the controls resemble each other.

These intervals need one caution. They describe what would change with other participants. The article very probably analysed the same participants as our file (same three group sizes), so sampling cannot be the cause of the difference between its equations and ours. The intervals tell us how much weight the equations can carry, not why the two analyses differ. For that we have to look inside the patient group.

1.7.3 Where the difference comes from: three patients

```
long_by_patient <- clean_results %>%
  filter(MovementTime >= 10) %>%
  count(SUBJ, name = "movements_over_10_s")

patients_with_ID3 <- Results %>%
  filter(GROUP == "pp", ID == "ID3") %>%
  distinct(SUBJ) %>%
  pull(SUBJ)

patients <- participant_lines %>%
  filter(GROUP == "pp") %>%
  left_join(long_by_patient, by = "SUBJ") %>%
  mutate(movements_over_10_s = replace_na(movements_over_10_s, 0),
         did_ID3 = ifelse(SUBJ %in% patients_with_ID3, "yes", "no")) %>%
  arrange(desc(slope))

patients %>%
  select(SUBJ, ID1, ID2, slope, intercept, movements_over_10_s, did_ID3) %>%
  knitr::kable(digits = 2,
               col.names = c("Patient", "MT ID1 (s)", "MT ID2 (s)",
                             "Slope", "Intercept", "Long mov.", "ID3"))
```

Patient	MT ID1 (s)	MT ID2 (s)	Slope	Intercept	Long mov.	ID3
ppS03	4.68	9.75	3.11	-7.15	21	no
ppS04	3.22	7.78	2.80	-7.40	8	no
ppS05	6.22	10.24	2.46	-3.14	25	no
ppS07	2.06	3.59	0.94	-1.51	1	no
ppS21	2.81	4.32	0.93	-0.73	1	yes
ppS29	1.94	3.41	0.90	-1.50	1	yes
ppS14	2.93	4.20	0.78	-0.03	1	yes
ppS23	2.02	3.16	0.70	-0.63	0	yes
ppS12	4.28	5.33	0.64	1.84	4	no
ppS09	2.92	3.94	0.63	0.53	0	no
ppS13	1.86	2.82	0.59	-0.39	0	yes
ppS10	5.24	6.11	0.53	3.21	2	no
ppS25	3.48	4.29	0.50	1.59	1	yes
ppS01	1.40	2.04	0.39	-0.09	0	yes
ppS20	3.01	3.61	0.37	1.60	0	yes
ppS16	1.25	1.70	0.28	0.18	0	yes
ppS22	1.20	1.65	0.27	0.17	0	yes
ppS11	5.03	5.33	0.19	4.32	5	no
ppS06	4.37	4.43	0.04	4.22	2	no

The slopes of the patients do not form one cloud around 0.9 s/bit. 16 patients have a slope between 0.04 and 0.94 s/bit, and 3 patients (ppS03, ppS04, ppS05) have a slope between 2.46 and 3.11 s/bit, with nothing in between. The median slope of the 19 patients is 0.63 s/bit, which is close to the article (0.672), whereas the mean is 0.9. Our group equation is a mean, and it is pulled by these 3 patients.

The table below shows how much. Below the table, the two numbers are the lowest and highest intercepts obtained when the equation is recomputed 19 times, each time without one patient.

```
line_of <- function(data) {
  summarise(data, intercept = mean(intercept), slope = mean(slope))
}

leave_one_out <- map_dfr(as.character(patients$SUBJ),
  \(s) line_of(filter(patients, SUBJ != s)))

bind_rows(
  "Article (Fig. 3)" = filter(article, GROUP == "pp") %>% select(intercept, slope),
  "Ours, 19 patients" = line_of(patients),
  "Ours, without the steepest patients" = line_of(others),
  .id = "equation"
) %>%
  mutate(MT_easy = intercept + slope * 3.80,
    MT_hard = intercept + slope * 5.43) %>%
  knitr::kable(digits = c(0, 3, 3, 2, 2),
```

```
col.names = c("Equation of the patients", "Intercept (s)", "Slope (s/bit)",
              "MT at 3.80 bits (s)", "MT at 5.43 bits (s)"))
```

Equation of the patients	Intercept (s)	Slope (s/bit)	MT at 3.80 bits (s)	MT at 5.43 bits (s)
Article (Fig. 3)	0.422	0.672	2.98	4.07
Ours, 19 patients	-0.259	0.898	3.15	4.62
Ours, without the steepest patients	0.798	0.543	2.86	3.75

```
range(leave_one_out$intercept)
```

```
[1] -0.5134201  0.1376385
```

Leaving out a single patient is enough to move the intercept from one side of zero to the other. Without the 3 steepest patients, the intercept is positive and the predicted movement times are close to those of the article, slightly below. We do not propose to remove these patients: the article did not (see the degrees of freedom above), and they are probably the patients for whom the task says the most. The point is that the equation of the patient group describes two kinds of patients with one line.

Our first idea had been to look at very long movements instead of participants. It leads to the same place: 72 movements last 10 s or more, all of them in the patient group, and 54 of them come from these 3 patients. The file also tells us what these movements are, through a column that we had not used so far:

```
clean_results %>%
  filter(GROUP == "pp") %>%
  group_by(duration = ifelse(MovementTime >= 10, "10 s or more", "under 10 s")) %>%
  summarise(movements = n(),
            median_peaks = median(NbVelPeaks))
```

```
# A tibble: 2 x 3
  duration      movements median_peaks
  <chr>          <int>         <dbl>
1 10 s or more      72           45
2 under 10 s     1448           7
```

The long movements have many more velocity peaks than the others. They are not recording errors with an impossible duration; they are movements made of many small sub-movements. So there is no reason in our data to discard them. In the article, movement time ends at “the first velocity minimum that was within the target” (Mottet et al. 2017, Data analysis). For a movement made of many sub-movements, where this end is placed depends on how the velocity signal is filtered and parsed, and a different choice could shorten these movements a lot, mainly for these patients and

at the difficult level. We only have the final values, so we cannot redo the parsing. But we can check whether our file was processed like the data of the article, by comparing group means with its Table 2. We assume that `NbVelPeaks` is what the article calls the number of sub-movements, which it counts as the number of velocity maxima.

```
table2 <- tibble(
  GROUP      = factor(c("pp", "ap", "cp"), levels = levels(clean_results$GROUP)),
  MT_article = c(3.53, 1.17, 0.92),
  nSM_article = c(7.52, 1.84, 1.99)
)

clean_results %>%
  group_by(GROUP) %>%
  summarise(MT_ours = mean(MovementTime),
            peaks_ours = mean(NbVelPeaks)) %>%
  left_join(table2, by = "GROUP") %>%
  select(GROUP, MT_ours, MT_article, peaks_ours, nSM_article) %>%
  knitr::kable(digits = 2,
               col.names = c("Group", "Mean MT, ours (s)", "Mean MT, article (s)",
                             "Velocity peaks, ours", "Sub-movements, article"))
```

Group	Mean MT, ours (s)	Mean MT, article (s)	Velocity peaks, ours	Sub-movements, article
pp	3.88	3.53	10.92	7.52
ap	1.13	1.17	2.39	1.84
cp	0.94	0.92	2.62	1.99

The movement times of the controls match the article within 0.04 s, and those of the patients are longer in our file. The number of velocity peaks is higher in our file for the three groups, and much higher for the patients. If the two columns measure the same thing, the movements in `Results.txt` were not parsed exactly as in the article. This does not prove that parsing explains the longer movement times of the patients, but it is the only explanation we found that is compatible with everything above: same participants, same number of movements, same way of computing the equations, and a difference that is concentrated in the most fragmented movements.

1.7.4 The patients who did not do ID3

The last column of the table of patients shows something we had treated as a simple gap in section 1.3. The patients who did not do the third level of difficulty are not a random half of the group:

```
patients %>%
  group_by(did_ID3) %>%
  summarise(patients = n(),
            MT_ID1 = mean(ID1),
            MT_ID2 = mean(ID2),
```

```

    steepest = sum(slope > 2)) %>%
knitr::kable(digits = 2,
              col.names = c("Did ID3", "Patients", "Mean MT at ID1 (s)",
                           "Mean MT at ID2 (s)", "Among the steepest"))

```

Did ID3	Patients	Mean MT at ID1 (s)	Mean MT at ID2 (s)	Among the steepest
no	9	4.22	6.28	3
yes	10	2.19	3.12	0

```
test_ID3 <- wilcox.test(ID2 ~ did_ID3, data = patients)
```

The patients without ID3 are about twice as slow on the two levels that everybody did (Wilcoxon rank-sum test on the movement time at ID2, $p < .001$), and the steepest patients are all among them. The simplest explanation is that the hardest level was not feasible, or not proposed, for the most affected patients.

We can look at this with Table 1 of the article, which gives for each patient the Enjalbert score, a prehension score from 0 (no motricity) to 6 (sub-normal). The patient numbers of the table (P01 to P29) are those of our file.

```

table1 <- tibble(
  SUBJ      = sprintf("ppS%02d", c(1, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14,
                                   16, 20, 21, 22, 23, 25, 29)),
  enjalbert = c(5, 3, 2, 4, 2, 4, 2, 3, 2, 3, 4, 5, 6, 2, 2, 3, 3, 2, 5)
)

patients_clinical <- patients %>%
  mutate(SUBJ = as.character(SUBJ)) %>%
  left_join(table1, by = "SUBJ")

stopifnot(!anyNA(patients_clinical$enjalbert)) # every patient found in Table 1

patients_clinical %>%
  group_by(did_ID3) %>%
  summarise(patients = n(), mean_enjalbert = mean(enjalbert))

```

```

# A tibble: 2 x 3
  did_ID3 patients mean_enjalbert
  <chr>      <int>      <dbl>
1 no         9         2.78
2 yes       10         3.7

```

```

cor_enjalbert <- cor.test(patients_clinical$enjalbert, patients_clinical$mean_MT,
                          method = "spearman")
test_enjalbert <- wilcox.test(enjalbert ~ did_ID3, data = patients_clinical)

```

The scores go in the expected direction but do not settle the question. Patients with a lower score tend to be slower (Spearman correlation between the score and the mean movement time: -0.42 , $p = 0.070$), and the patients without ID3 have a lower score on average, but this difference could be due to chance ($p = 0.199$). The 3 steepest patients have scores of 3, 2, 4, in the middle of the scale. So the patients without ID3 are clearly the slower ones on this task, but we cannot say that they are the more impaired ones clinically. The article gives the same limit: there was no systematic assessment of the severity of the impairment.

This changes how we see our decision of section 1.3. Removing ID3 was not only a matter of having complete cells. An analysis that kept ID3 with the patients who did it would have described the faster half of the patients and presented it as the patient group.

1.7.5 More sensitive to difficulty: in seconds or in proportion?

Our sentence in section 1.3 says that difficulty costs more to the patients. In seconds it does. But the patients are also about three to four times slower at the easy level, and a group that is slower by a constant factor would have a steeper line without being more sensitive to difficulty. To separate the two, we look at the ratio of the two movement times of each participant.

```
participant_lines %>%
  group_by(GROUP) %>%
  summarise(difference_s = mean(difference),
            ratio = mean(ratio)) %>%
  knitr::kable(digits = 2,
               col.names = c("Group", "MT at ID2 minus MT at ID1 (s)",
                             "MT at ID2 divided by MT at ID1"))
```

Group	MT at ID2 minus MT at ID1 (s)	MT at ID2 divided by MT at ID1
pp	1.46	1.48
ap	0.27	1.27
cp	0.20	1.25

```
test_ratio <- kruskal.test(ratio ~ GROUP, data = participant_lines)
```

Going from the easy to the difficult level adds about a quarter to the movement time of the controls and about a half to that of the patients. The difference between groups is still there (Kruskal-Wallis test on the ratios, $p = 0.027$), but it is much less clear than in seconds, and it would not pass the threshold of .01 used in the article. “Five to seven times steeper” and “about twice the relative cost” describe the same data. Most of the difference between the slopes comes from the fact that the patients are slower everywhere.

1.7.6 Checking the ANOVA without assuming equal variances

In section 1.3 we saw that the standard deviation between participants is about 22 times larger for the patients than for the young controls, which the ANOVA does not allow. With two levels

of ID the check is simple, because the two effects that involve GROUP reduce to one number per participant: the main effect of GROUP compares the participants' mean movement time (`mean_MT`), and the `GROUP × ID` interaction compares their difference between the two levels (`difference`). We test both with rank tests, which make no assumption about variances.

```
test_group      <- kruskal.test(mean_MT ~ GROUP, data = participant_lines)
test_interaction <- kruskal.test(difference ~ GROUP, data = participant_lines)

# Which groups differ? Wilcoxon tests two by two, Holm correction
pairs_group <- pairwise.wilcox.test(participant_lines$mean_MT,
                                     participant_lines$GROUP,
                                     p.adjust.method = "holm")
pairs_interaction <- pairwise.wilcox.test(participant_lines$difference,
                                           participant_lines$GROUP,
                                           p.adjust.method = "holm")

pairs_group$p.value
```

	pp	ap
ap	8.060486e-08	NA
cp	1.697540e-10	0.002594354

```
pairs_interaction$p.value
```

	pp	ap
ap	1.809003e-05	NA
cp	5.941390e-07	0.07912687

Both effects are confirmed (Kruskal-Wallis, GROUP: $p < .001$; interaction: $p < .001$), so the conclusion of section 1.3 does not depend on the assumption that was violated.

The comparisons two by two give the post-hoc information that was missing in section 1.3. For the effect of difficulty (second table), the patients differ from both control groups and the two control groups do not differ from each other, as in the article. For the overall movement time (first table), the patients again differ from both control groups, but the two control groups also differ from each other: the age-matched controls are about 0.2 s slower than the young ones. On the graph of section 1.5 the two control lines look identical only because the axis is scaled for the patients.

The article reports that the two control groups do not differ (Tukey contrasts, $p = 0.79$). We find the same when we use the same kind of test:

```
TukeyHSD(aov(mean_MT ~ GROUP, data = participant_lines))$GROUP
```

	diff	lwr	upr	p adj
ap-pp	-2.7510670	-3.751662	-1.7504720	7.560454e-08
cp-pp	-2.9492914	-3.851216	-2.0473674	9.026866e-10
cp-ap	-0.1982245	-1.198819	0.8023704	8.814699e-01

The two results do not contradict each other. Tukey’s test compares each difference with a variability pooled over the three groups, and here this variability is mostly that of the patients. A difference of 0.2 s between controls is small compared with it, although it is large compared with the variability of the controls themselves. The rank test does not pool the variances. That the age-matched controls are a little slower than the young ones is therefore likely, but it is a small effect next to the effect of stroke.

These tests were chosen after looking at the data, so they are checks of our first analysis, not new confirmatory results.

1.7.7 Are the conclusions similar?

For what the article concludes, yes. The three effects of the ANOVA are significant in both analyses, the F values of GROUP and of the interaction are of the same order, rank tests confirm them, and the patients differ from both control groups. The movement times of the controls agree with the article within 0.1 s, and the intercepts do not differ between groups in either analysis.

For the equation of the patients, the question is not well posed. Our intercept has the opposite sign, but its confidence interval covers more than a second on each side of zero and one patient decides the sign. The article is careful on the same point: its intercepts do not differ between groups and the authors interpret this “with caution, because extrapolations are very sensitive to errors” (Mottet et al. 2017, Discussion). The difference that deserves attention is the slope, and it comes from 3 patients whose long, fragmented movements are longer in our file than they seem to be in the data behind the published figure.

Two things in our data go beyond the comparison. The statement that patients are five to seven times more sensitive to difficulty holds in seconds; in proportion to their own speed it is about twice. And “the patients” is not one homogeneous group in this file: the slower half did not do the third level, and three of them account for most of the difference between our equation and the published one.

Data sources

File	Description	Source (URL)
data/Results.txt	One row per reaching movement: GROUP, SUBJ, TRIAL, ORI, DIR, ID, REP, NbVelPeaks, MovementTime	Course Moodle page, IEAP Series 03: https://moodle.umontpellier.fr/mod/folder/view.php?id=1025707
Fig. 3 of the article	Reference figure and equations (MT as a function of ID, by group)	https://doi.org/10.1371/journal.pone.0173674

Software

Analyses were run in R (R version 4.6.1 (2026-06-24)) with the packages `ez` (4.5.0), `tidyverse` (2.0.0) and `here` (1.0.2). The report was written with Quarto.

References

Mottet, Denis, Liesjet Elisabeth Henriette van Dokkum, Jérôme Froger, Abdelkader Gouaïch, and Isabelle Laffont. 2017. “Trajectory Formation Principles Are the Same After Mild or Moderate Stroke.” *PLOS ONE* 12 (3): e0173674. <https://doi.org/10.1371/journal.pone.0173674>.